

Experiment 19: Triple DES in CBC Mode:

```
from Crypto.Cipher import DES3
from Crypto.Util.Padding import pad, unpad

key = b"Sixteen byte key"
iv = b"12345678"

pt = input("Enter plaintext: ").encode()

cipher = DES3.new(key, DES3.MODE_CBC, iv)
ct = cipher.encrypt(pad(pt, 8))

print("Ciphertext:", ct.hex())

decipher = DES3.new(key, DES3.MODE_CBC, iv)
dt = unpad(decipher.decrypt(ct), 8)

print("Decrypted Text:", dt.decode())
```

Output:

```
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Enter Plain Text: HELLO
Cipher Text: 62d4d1f294619f42
Plain Text: HELLO
|
```